

BM
4/28/22

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES

DATE COLLECTED: 18 APR 2022 Initials: TW

Conversion Factors

10 in. Bulk Rain Funnel: NET divided by 51.1 = catch

N/A if Overfilled

12 in. Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	CATCH (mm)	ETI CATCH (mm)	Coll EFF (% of ETI)
RG-22	<u>0735</u>	<u>1027</u>	<u>20.1</u>		
RG-1	<u>1040</u>	<u>1215</u>	<u>23.8</u>		
RG-4	<u>1000</u>	<u>1021</u>	<u>20.0</u>		
RG-23	<u>0850</u>	<u>1143</u>	<u>22.4</u>		

Volumes Collected				
	RG-22	RG-1	RG-4	RG-23
Gross g	<u>1196</u>	<u>1382</u>	<u>1189</u>	<u>1312</u>
Tare g	<u>169</u>	<u>167</u>	<u>168</u>	<u>169</u>
Net g	<u>1027</u>	<u>1215</u>	<u>1021</u>	<u>1143</u>

SITE	CONTAMINATION					GB Ring	dBASE FIELD code?	SAMPLE REJECTED @LEVEL =	Database CODES checked ?
	(0 = none; 1 = low Fine PM)	(0 = none; 1 = low Coarse PM)	(0 = none; 1 = low Bird FECES)	(0 = none; 1 = low Insects)	(0 = none; 1 = low Color)				
RG-22	<u>2</u>	<u>2</u> ↑	<u>0</u>	<u>1</u> Tiny diptera	<u>0</u>	<u>-</u>			
RG-1	<u>1</u>	<u>1</u> Bark seeds	<u>0</u>	<u>2</u> Big diptera	<u>0</u>	<u>-</u>			
RG-4	<u>2</u>	<u>1</u> ↓	<u>0</u>	<u>1</u> Colembola	<u>0</u>	<u>-</u>			
RG-23	<u>1</u>	<u>1</u> yb seed	<u>0</u>	<u>0</u>	<u>0</u>	<u>-</u>			

Precipitation amount
0.87 inches / 22.2 mm
ETI collector

ADDITIONAL COMMENTS: First funnel/bottle collection. Precip was mostly rain.
some snow fell @ end of 4/16 storm @ high. elev. RG-23 funnel had snow
@ collection, dumped it in bottle, but some was lost.

Applicable dBASE FIELD codes.
(See SOP Manual)

970 = Obvious Contamination: Sample REJECTED

969 = Possible Contamination: Check Return Analyses

955 = QNS Quantity Not Sufficient for analyses (< 100 mL NET catch)

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

Streamwater Samples

Date Collected.

18 APR 2022

Initials:

Tw

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none:1 = low---->5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	<u>1050</u>	<u>6.7°</u>	<u>.279</u>	<u>1</u>	<u>1-</u>	<u>1</u>	<u>Algae below v-notch</u>	
WS-2	<u>1030</u>	<u>6.0°</u>	<u>.259</u>	<u>L</u>	<u>0</u>	<u>0</u>	<u>Algae v-notch wall</u>	
WS-3	<u>1015</u>	<u>4.6°</u>	<u>.301</u>	<u>L</u>	<u>1-</u>	<u>1</u>	<u>Thick algae.</u>	
WS-4	<u>1100</u>	<u>6.5°</u>	<u>.378</u>	<u>L</u>	<u>0</u>	<u>1</u>	<u>Algae-</u>	
WS-5	<u>1210</u>	<u>7.5°</u>	<u>.396</u>	<u>L</u>	<u>0</u>	<u>0</u>		
WS-6	<u>1130</u>	<u>7.3°</u>	<u>.306</u>	<u>L</u>	<u>1-</u>	<u>1</u>	<u>Algae</u>	
WS-7	<u>0920</u>	<u>2.0°</u>	<u>.592</u>	<u>L</u>	<u>1-</u>		<u>splash ice, trace snow in watershed</u>	
WS-8	<u>0910</u>	<u>1.8°</u>	<u>.487</u>	<u>L</u>	<u>1</u>	<u>2</u>	<u>glaze, splash ice, "</u>	
WS-9	<u>0940</u>	<u>0.6°</u>	<u>.477</u>	<u>L</u>	<u>2</u>	<u>0</u>	<u>Weir frozen heavy splash ice in</u>	
Hubbard BK	<u>1235</u>	<u>4.5°</u>		<u>F</u>	<u>1</u>	<u>1</u>	<u>stream. Frozen flatsam, 25-50% snow</u>	
Mlake Outlet	<u>1250</u>	<u>8.1°</u>		<u>L</u>	<u>1-</u>	<u>0</u>	<u>OTB</u>	

ADDITIONAL COMMENTS: Elevated flows, cool wet weather continues NE watersheds had newly
glazed rocks, splash ice (0-50% ^{watershed}), Dusting fresh snow, some melted out.
No birds singing yet, woods are quiet!

Applicable dBASE FIELD codes: 911= High Flow Event in Progress; 912 = Drought Conditions (syringe from Standing Pool), 969 = possible problems (note)

955 = QNS. Quantity Not Sufficient for analyses

master sample notes 2017

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 19 APR 2022RUN By: TW

Meter CALIBRATION: See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.14 xat: 17.6 °C Lab Temp

Samples:	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Precipitation					
STATION	<u>18 APR 2022</u>	<u>5.56</u>	<u>5.61</u>	<u>5.7</u>	
RG-1	<u>↓</u>	<u>5.51</u>	<u>5.51</u>	<u>6.0</u>	
RG-4	<u>↓</u>	<u>5.29</u>	<u>5.26</u>	<u>7.0</u>	
RG-23	<u>18 APR 2022</u>	<u>5.27</u>	<u>5.29</u>	<u>7.7</u>	

		Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Streamwater							
WS-1	<u>18 APR 2022</u>	<u>5.62</u>	<u>+6.3</u>	<u>5.19</u>	<u>5.60</u>	<u>11.8</u>	
WS-2		<u>5.56</u>	<u>+5.7</u>	<u>5.23</u>	<u>5.60</u>	<u>8.1</u>	
WS-3		<u>5.75</u>	<u>+8.4</u>	<u>5.46</u>	<u>5.79</u>	<u>9.0</u>	
WS-4		<u>6.05</u>	<u>+16.6</u>	<u>5.22</u>	<u>6.06</u>	<u>9.6</u>	
WS-5		<u>5.48</u>	<u>+3.6</u>	<u>5.20</u>	<u>5.43</u>	<u>9.5</u>	
WS-6		<u>5.53</u>	<u>+0.2</u>	<u>5.44</u>	<u>5.57</u>	<u>9.1</u>	
WS-7		<u>5.76</u>	<u>+12.5</u>	<u>5.29</u>	<u>5.78</u>	<u>8.6</u>	
WS-8		<u>5.47</u>	<u>+3.6</u>	<u>5.24</u>	<u>5.47</u>	<u>9.0</u>	
WS-9		<u>4.71</u>	<u>-24.2</u>	<u>4.47</u>	<u>4.75</u>	<u>13.1</u>	
Hubbard BK	<u>↓</u>	<u>6.13</u>	<u>+22.9</u>	<u>5.27</u>	<u>6.16</u>	<u>9.8</u>	
Mlake Outlet	<u>18 APR 2022</u>	<u>6.50</u>	<u>+68.1</u>	<u>5.06</u>	<u>6.45</u>	<u>32.1</u>	

ADDITIONAL COMMENTS: Saturated conditions, snowmelt chem/NF

Applicable codes: 911= High Flow Event in Progress;

955= QNS. Quantity Not Sufficient for analyses(<100 mL NET catch)

912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

B/M

DIC-[date].xls

<-save as [date]= mm-dd-yy

HBEF Stream and M.Lake Routine Sample DIC Datadate: **19APR2022**

Samples collected 04/18/2022

Aqueous STANDARDS

type	DIC	peak ht	adjusted	ATT -X
	set =	mm	mm	
DIW blank	0	11	0	16
	0	11	0	16
	0	12	0	16
100 µM	100	30	19	16
	100	31	20	16
	100	31	19	16
200 µM	200	53	42	16
	200	52	41	16
	200	53	41	16
400 µM	400	100	89	16
	400	98	87	16
	400	99	87	16

Statistics

Values from matrix at M5 -->

slope = 4.524
 slope ± 0.060
 Y-int = 7.2
 Y-int ± 3.0
 R-sq = 0.9982

Laboratory Conditions

Lab Temp = 18.5 °C
 Barom Pressure = 756 mm Hg
 Analyst's Initials TW 60%RH
 NaCO3 Standards:
 0.1M Stock Dated : 3/10/22
 0.0 M Working Solution Dated : 3/10/22

Samples

	inj 1	inj 2	inj 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC	Calc	NOTES
	1	2	3					µM/L	err±	
ML-70	29	31	30	16	30.0	1.0	ML-70	136	5	120 mA current
	0	0	0	16	0.0	0.0	0	0	0	
	0	0	0	16	0.0	0.0	0	0	0	
	0	0	0	16	0.0	0.0	0	0	0	
	0	0	0	16	0.0	0.0	0	0	0	
W-1	18	17	17	16	17.3	0.6	W-1	78	3	2 inj
W-2	12	13	14	16	13.0	1.0	W-2	59	5	
W-3	11	12	13	16	12.0	1.0	W-3	54	5	
W-4	13	12	12	16	12.3	0.6	W-4	56	3	
W-5	10	9	11	16	10.0	1.0	W-5	45	5	
W-6	9	10	10	16	9.7	0.6	W-6	44	3	
W-7	11	11	12	16	11.3	0.6	W-7	51	3	
W-8	10	10	10	16	10.0	0.0	W-8	45	0	
W-9	11	11	11	16	11.0	0.0	W-9	50	0	
HBk	12	12	13	16	12.3	0.6	HBk	56	3	
				16	0.0	0.0	0			
Lab Air	21	21	21	16	21.0	0.0	Lab Air	95	0	
Outside Air	12	12	11	16	11.7	0.6	Outside Air	53	3	
Gas 1000	24	24	25	16	24.3	0.6	Gas 1000	110	3	std=83.3
Gas 3000	73	75	75	16	74.3	1.2	Gas 3000	336	5	std=246.3